

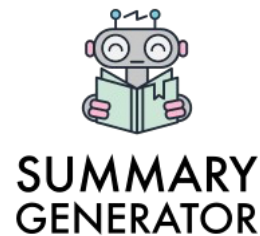
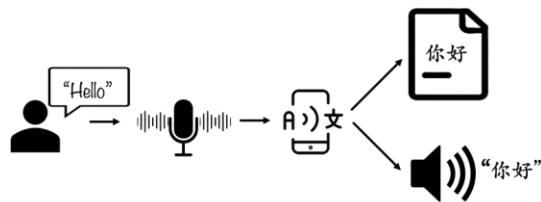
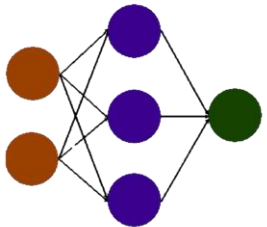
RETHINKING ATTENTION WITH PERFORMERS

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Transformers

What are Transformers?

- ▶ Transformers are powerful neural network architectures.
- ▶ They are widely used for many NLP tasks such as speech recognition and machine translation.
- ▶ Text generation, summarization, and time-series prediction are some other examples.
- ▶ They have significant impact in sequence modeling.



Attention Mechanism

- ▶ Transformers use Attention Mechanism to capture complex dependencies.
- ▶ They scale quadratically with sequence length L . So, computational cost limits Transformers in low-resource or long-sequence settings.
- ▶ Many methods attempt to fix this by restricting or structural priors.
- ▶ Some methods use low-rank kernel approximations to replace softmax attention with more efficient dot-product feature maps.
- ▶ These kernel-based methods depend on kernels that can be expressed as dot-products of finite positive feature vectors.

Are they logical?

- ▶ Many proposed methods do not approximate true attention, just replace it with simpler mechanisms.
- ▶ These alternatives often impose limitations on deeper networks.
- ▶ There are no strong theoretical guarantees for their representation power.
- ▶ NOT LOGICAL but What is the weakness?



Weakness

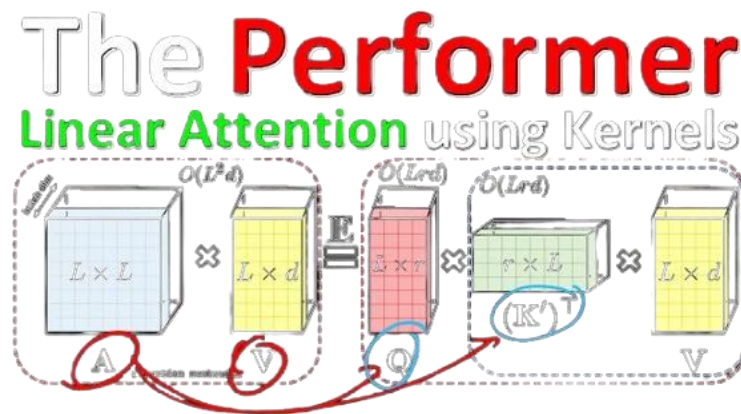
- ▶ Validating sparsity patterns → It requires manual GPU kernel engineering.
- ▶ Some approaches reduce memory, but these may still fail on long-sequence tasks.
- ▶ Shortened backpropagation methods cannot capture long-range dependencies.
- ▶ Some biased-attention approximations work only in non-causal settings and it causes exhibit large estimation error.



Performer

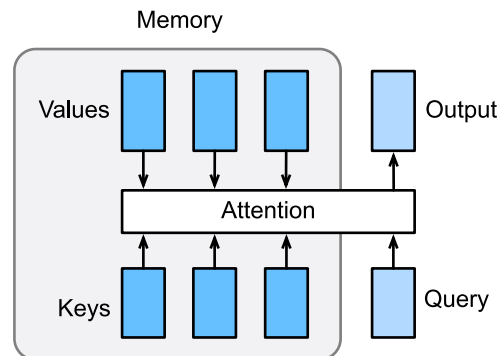
FAVOR+ mechanism & PORFs

- ▶ Performer is a model which can estimate Transformers full-rank-attention softmax with using only linear space and time complexity
- ▶ BIG POINT: Performer does not rely on any priors.
- ▶ Performers use a novel Fast Attention Via Positive Orthogonal Random Features approach



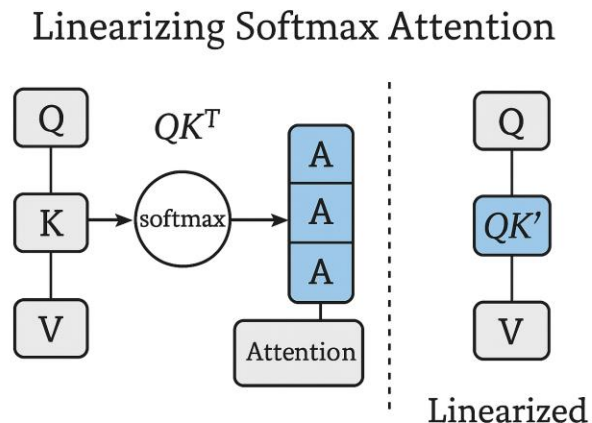
Regular attention mechanism

- ▶ Standard Transformer are computationally expensive.
- ▶ Each of Q (queries), K (keys), V (values) has L rows.
- ▶ For attention matrix we have $\text{Similarity} = QK^T$, $A = \text{softmax}(QK^T)$ and $\text{Att}(Q, K, V) = AV$.
- ▶ The main issue: Computing and storing the full $L \times L$ attention matrix requires $O(L^2)$ time and memory.
- ▶ How to solve? **The only way to break the quadratic bottleneck is to rewrite or approximate the softmax kernel in a linear form.**



Generalized Kernel Attention

- ▶ It is the idea behind linearizing Softmax attention.
- ▶ Instead of viewing attention as something specific, the paper generalize it using $(A=D^{-1}K(Q, K))$ where K is any valid kernel function applied to queries and keys, and D is a normalization matrix.
- ▶ A kernel K is called kernelizable if it can be written exactly or approximately as an inner product in some feature space.



- ▶ This representation is crucial because it allows attention to be computed in linear time.
- ▶ Softmax is only one example of such a kernel, but many other kernels—Gaussian, Laplacian, and other dot-product kernels—also fit into this framework.
- ▶ Performers are not just a faster approximation of softmax attention; they also provide a general mathematical framework.

How To Approximate Softmax Kernels

Many prior softmax approximations are invalid:

- ▶ Standard RFF (cos/sin) → negative outputs, high variance
- ▶ Biased estimators → low accuracy, only usable in non-causal settings
- ▶ Non-positive kernels → unstable attention weights

The correct approach:

- ▶ Rewrite Softmax as an expectation of a positive kernel
- ▶ Use Positive Random Features (PRF) to ensure (I) Unbiased estimation (II) Strictly positive feature maps (III) Low variance.

Orthogonal Random Features

- ▶ ORFs can be applied to reduce the variance of Softmax/Gaussian kernel estimators for any dimensionality.
- ▶ Also leads to the first exponentially small bounds on large deviations probabilities that are strictly smaller than non-orthogonal methods.

Results

What we get?

- ▶ Approximate softmax kernels with low-variance, when the kernel value ($SM(x,y)$) is very small, gets big variance and it will cause to error.
- ▶ Main reason is they mosly use in attention matrix.
- ▶ So, The hyperbolic version is more accurate.

$$\text{MSE}(\widehat{SM}_m^{\text{hyp}+}(\mathbf{x}, \mathbf{y})) = \frac{1}{2}(1 - \exp(-\|\mathbf{x} + \mathbf{y}\|^2))\text{MSE}(\widehat{SM}_m^+(\mathbf{x}, \mathbf{y})),$$

What are the proofs?

Theorem 1

- ▶ The regularized softmax kernel used by Positive Random Features (PRFs) is always slightly smaller than the true softmax kernel, but never much smaller.
- ▶ In fact, the ratio ($\mathbf{A}_{\text{reg}}(i,j) / \mathbf{A}(i,j)$) stays extremely close to 1, within a difference which is a very small number.

$$\inf_{i,j} \frac{\mathbf{A}^{\text{reg}}(i,j)}{\mathbf{A}(i,j)} \geq 1 - \frac{2}{d^{\frac{1}{3}}} + o\left(\frac{1}{d^{\frac{1}{3}}}\right), \text{ and } \sup_{i,j} \frac{\mathbf{A}^{\text{reg}}(i,j)}{\mathbf{A}(i,j)} \leq 1.$$

Theorem 2

- ▶ Orthogonal Random Features (ORFs) always reduce the variance compared to Standard Random Features.
- ▶ Formally, the Mean Square Error (MSE) of the orthogonal estimator is always smaller by a strictly positive amount for any dimension.
- ▶ What they mean:
 - ORFs provide more stable estimates and give lower variance
 - They are always better than non-orthogonal features
 - This holds for all practical dimensions.

Theorem 3

- ▶ Gives exponential tail bounds for the random-feature approximation of the regularized Softmax kernel.
- ▶ Two bounds are given: positive random features and positive orthogonal random features (ORFs).
- ▶ Result: The ORF bound is always strictly tighter.
- ▶ Meaning: ORFs make the softmax-kernel approximation far more *stable* and *safe*, especially in the small-kernel where attention is most sensitive.

$$\begin{aligned} \mathbb{P}[\widehat{\text{SMREG}}_m^+(\mathbf{x}, \mathbf{y}) > a] &\leq \exp(-\theta ma) M_Z(\theta)^m, \quad \mathbb{P}[\widehat{\text{SMREG}}_m^{\text{ort}+}(\mathbf{x}, \mathbf{y}) > a] \\ &\leq \exp(-\theta ma) \left(M_Z(\theta)^m - \exp\left(-\frac{m}{2}(\|\mathbf{x}\|^2 + \|\mathbf{y}\|^2)\right) \frac{\theta^4 m(m-1)}{4(d+2)} \|\mathbf{x} + \mathbf{y}\|^4 \right) \end{aligned}$$

Theorem 4

- ▶ FAVOR+ can approximate the entire attention matrix with a uniformly bounded error, using only a small number of random features.
- ▶ The required number of features does not depend on the sequence length. It only depends on the embedding dimension.
- ▶ FAVOR+ can therefore scale to very long sequences while keeping the error controlled.

Experiments

Implementation and Comparison

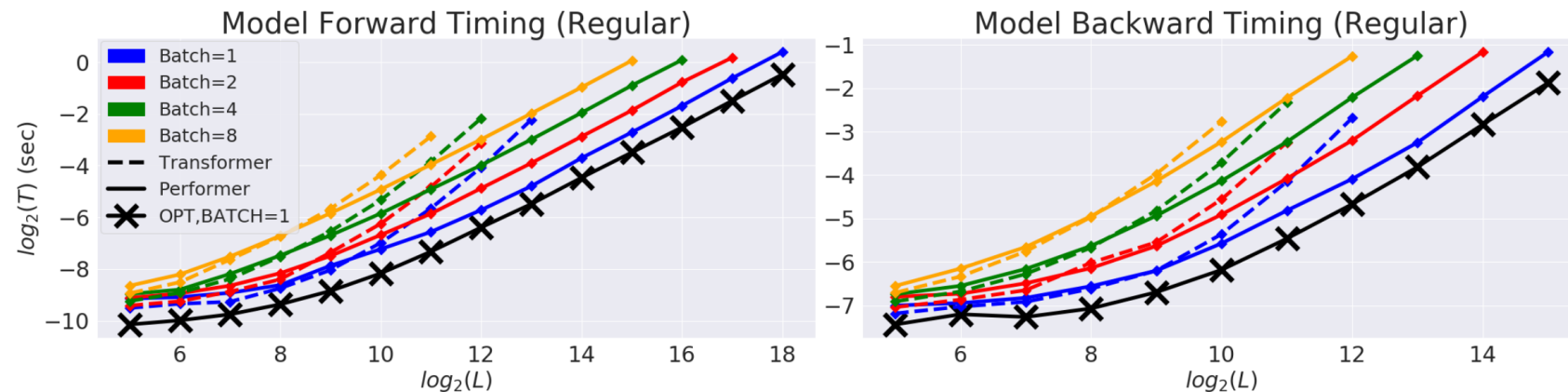
- ▶ Performer components are same as Transformer. It only replace the Transformers Attention component.
- ▶ They use Transformer, Unidirectional Reformer, Bidirectional Linformer for comparison.



COMPARISON

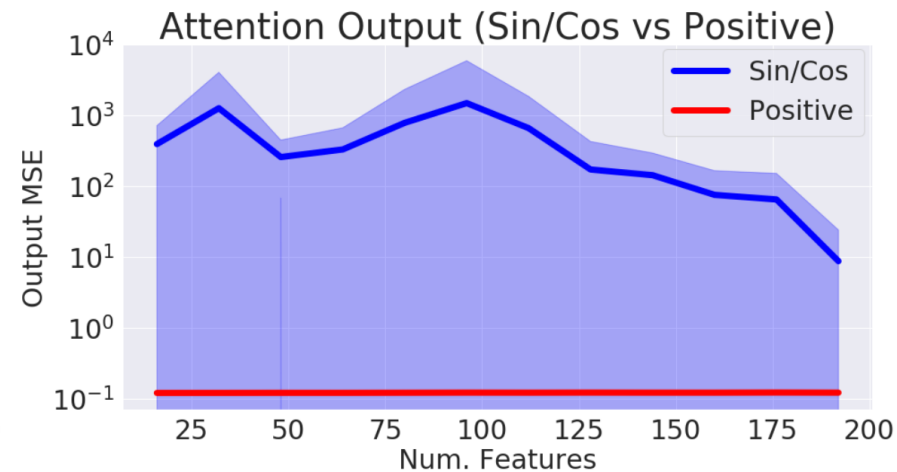
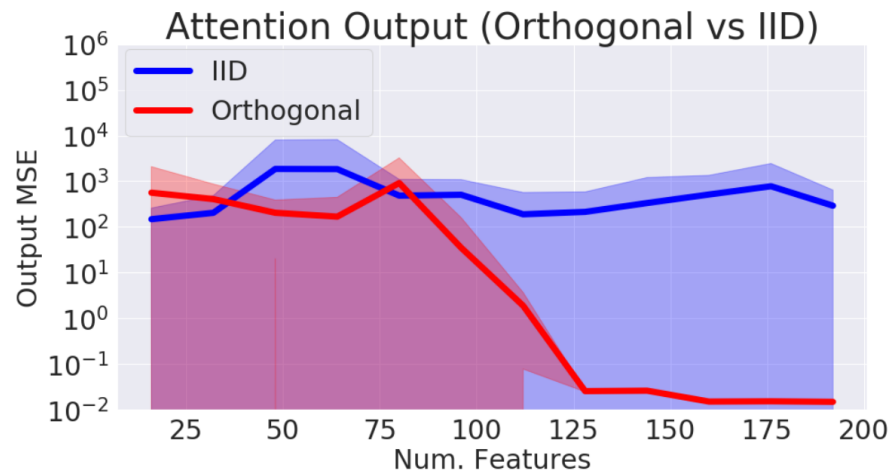
Computation Cost

- ▶ One of the computational Bottlenecks is Bidirectional Performer and Transformer Backward
- ▶ Using regular default size (N_{heads} , N_{layers} , MPL width, d) = (8, 6, 2048, 512)



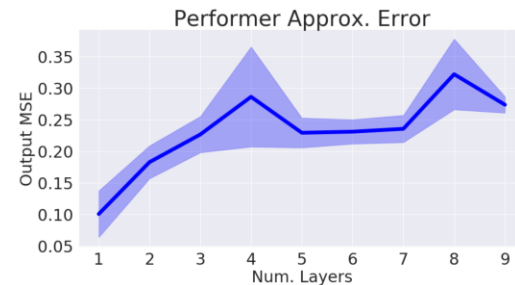
Softmax Attention Approximate Error

- ▶ Orthogonal features produce lower error than unstructured (IID) features.
- ▶ Positive features produce lower error than trigonometric features.
- ▶ Result: Orthogonal features and Positive features are valid for the PORF mechanism.

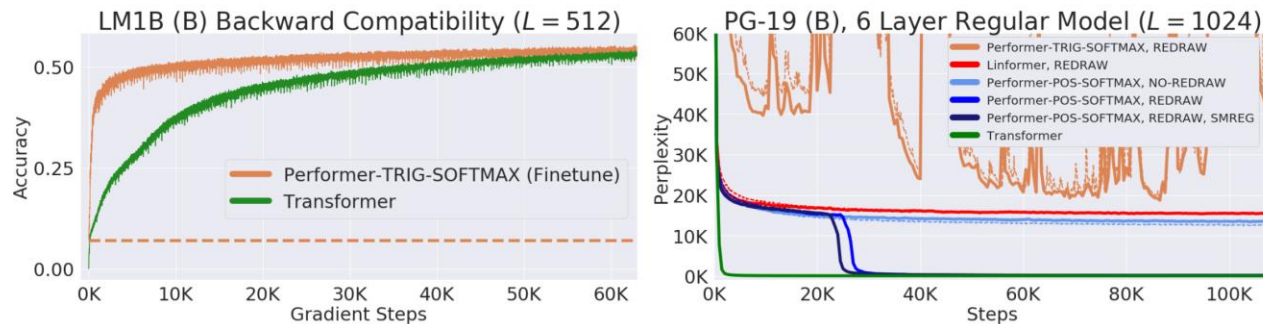


Softmax Approximate on Transformers

- ▶ Even the approximation of the attention mechanism is tight, small errors can easily propagate throughout multiple Transformer layers.

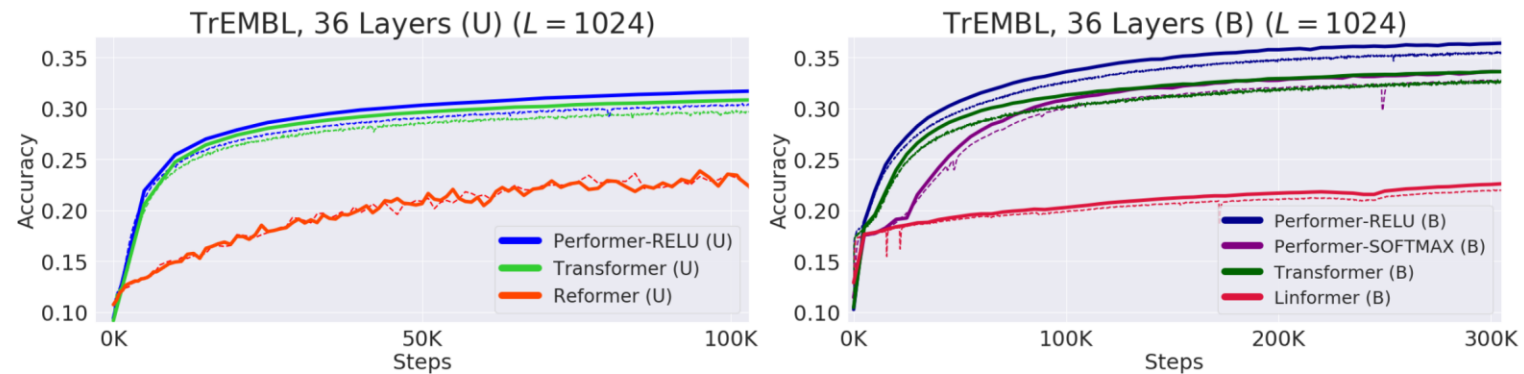


- ▶ By using small datasets, Attention models works well but for bigger datasets, we should use positive features



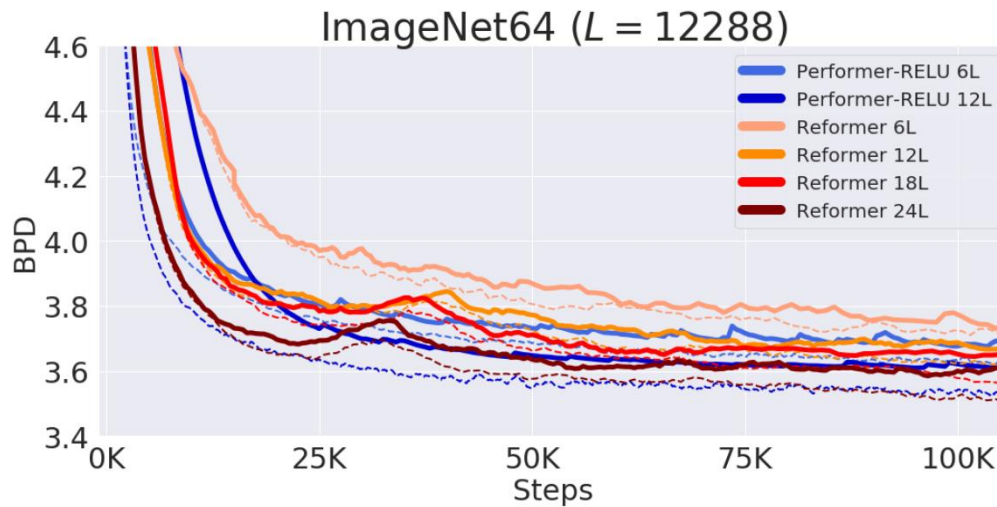
Multiple Layer Training

- ▶ Here is benchmark of both Unidirectional and Bidirectional Performer.
- ▶ It should be mentioned that the generalized attention by Performer-RELU achieves the highest accuracy in both cases.



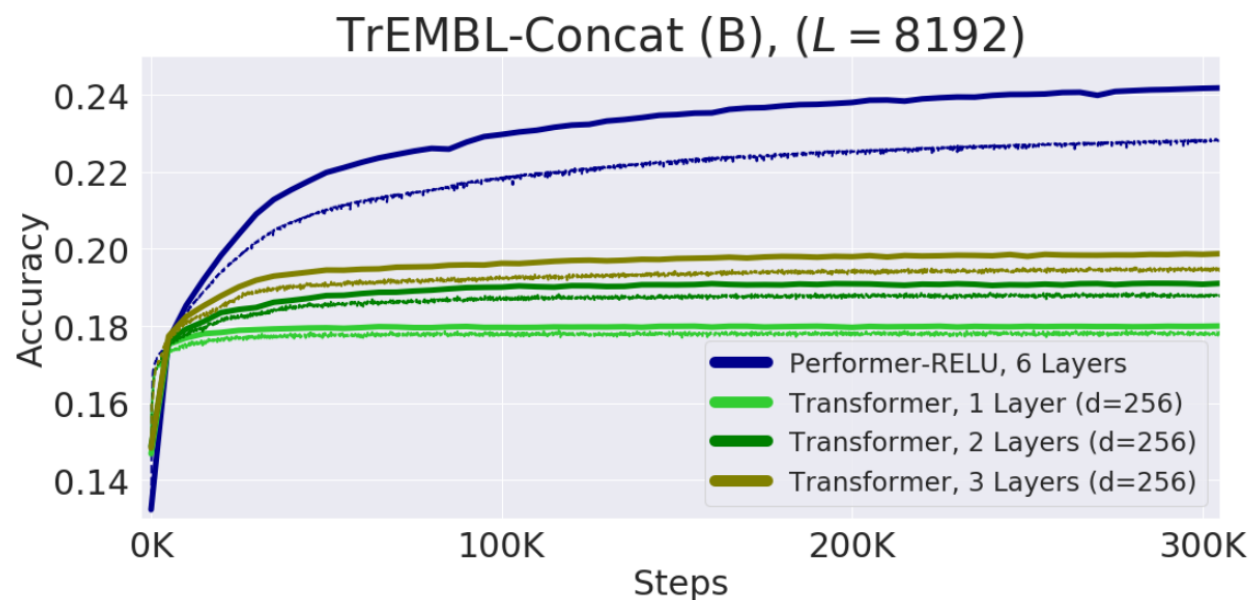
Large length Training

- ▶ Results for Performer-6 matches Reformer-12
- ▶ Depends on hardware, Performer is two times faster than Reformer.



Proof of Principle Study

- ▶ Created initial protein benchmark with a long length.
- ▶ A regular Transformer overloads memory even at a batch size of 1 per chip



Conclusion

Conclusion and usage

- ▶ Performer is a new type of Transformer, relying on Fast Attention Via positive Orthogonal Random features mechanism.
- ▶ This mechanism provides the first effective unbiased estimation of the softmax-based Transformer with linear space and time complexity.
- ▶ Some examples for usage is Biology and Medicine, Environment, Research on Transformers and etc.

Thanks for your attention

Any Question?